Solution to Ramanujan Challenge Problem 2.1: Polynomial Continued Fraction for π — Near-Complete Proof

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Abstract

We give a near-complete proof that the polynomial continued fraction in Problem 2.1 equals $6/(3-\pi)$. The proof reduces the order-3 carrier to a forced order-2 equation via the adjoint solution $z(x) = x^{-9/5}(x+2)$, linearizes the order-2 factor on the elliptic curve $E : u^2 = x(1+11x-x^2)$ using a third-kind differential with 3-torsion residues, and evaluates the endpoint coordinate as $T = -i\pi/(3\sqrt{3})$ via the logarithmic jump $\log(-1) = i\pi$. The remaining step is the algebraic de Rham reduction of the variation-of-parameters differentials.

1 Setup and factorization

With $\varphi = (1+\sqrt{5})/2$, $\alpha = \varphi^5$, $\delta = \varphi^{-5}$, $\beta = -\delta$, and $f(x) = 1 + 11x - x^2 = (\alpha - x)(x+\delta)$, the order-3 carrier is

$$\frac{\mathcal{L}_3}{20x^3 f(x)} = \left(D + \frac{6}{5x} + \frac{1}{x+2} + \frac{f'}{f}\right)\mathcal{L}_2,$$

where $\mathcal{L}_2$ is an explicit order-2 operator. The inhomogeneous equation is $\mathcal{L}_3 F_Q = 6x$.

2 Adjoint quadrature (algebraic step)

The adjoint solution $z(x) = x^{-9/5}(x+2)$ performs the first reduction exactly:

$$\mathcal{L}_2 F_Q = \frac{x+12}{4x(x+2)f(x)}.$$

This is **algebraic** — no π yet.

3 Elliptic linearization

On $E : u^2 = xf(x)$, the third-kind differential

$$\omega = \frac{x+2}{(x-3)u}\,dx$$

has poles at $P_\pm = (3, \pm 5\sqrt{3})$ with residues $\pm 1/\sqrt{3}$. Zero residue at the branch points.

The torsion relation: $P_+ - P_-$ has **order 3** in $\mathrm{Pic}^0(E)$, certified by the rational function

$$\mathcal{R}(x,u) = \frac{(\sqrt{3}\,u - 4x - 3)(3u - 5\sqrt{3}\,x)}{(\sqrt{3}\,u + 4x + 3)(3u + 5\sqrt{3}\,x)}, \quad d\log\mathcal{R} = 3\sqrt{3}\,\omega.$$

4 The π-producing integral

Define $t(x) = \int_0^x \omega$. Along the path from $(0,0)$ to $(\beta,0)$ through the upper half-plane:

$$\boxed{T := t(\beta) = -\frac{i\pi}{3\sqrt{3}},}$$

because $\mathcal{R}(\beta,0) = 1$, $\mathcal{R}(0,0) = -1$, and $3\sqrt{3}\,T = \log(1) - \log(-1) = -i\pi$.

With $a = \sqrt{3}/2$: $aT = -i\pi/6$, so $e^{\pm aT} = e^{\mp i\pi/6}$ are **sixth roots of unity**.

5 Connection to the continued fraction

The linearization $F = h(x)\,\psi(t(x))$ with $h = \sqrt{(x-3)/u}$ reduces $\mathcal{L}_2$ to $\psi'' - \frac{3}{4}\psi = 0$ (constant-coefficient!).

The CF value satisfies

$$\frac{6}{3-\pi} = -42 + 396 \cdot \frac{\psi_S(T)}{\psi_Q(T)},$$

where ψ_Q, ψ_S are the forced solutions corresponding to Q_n and $S_n = (P_n + 42Q_n)/396$.

6 Remaining step

The proof is complete modulo one finite algebraic computation: the de Rham reduction of the four variation-of-parameters differentials (two for ψ_Q, two for ψ_S) along the path from 0 to β. After this reduction, the only surviving transcendental is the torsion logarithm T, and the arithmetic simplifies to $6/(3-\pi)$.

References

[1] Ramanujan Machine Group, *The Ramanujan Challenge for AI*, 2026.